W8: Hybrid FELsim/RF-Track Stage 11 Optimisation

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1 Objective

Optimise Stage 11 of the UH MkV FEL beamline using RF-Track particle tracking instead of FELsim's linear transfer matrices. The goal is twofold: (a) validate the transfer matrix model against full nonlinear particle tracking, and (b) determine whether RF-Track can achieve a better undulator Twiss match by capturing effects absent from the linear model (space charge, higher-order optics, realistic dipole transport).

This study complements the COSY cross-validation (W4), which compared gradient-based and gradient-free *matrix* optimisers. Here we compare a *matrix* optimiser (FELsim) against a *particle tracking* optimiser (RF-Track) on the same problem.

2 Method

2.1 Hybrid Approach

Stages 1–10 are fast, local matches (1–4 variables each, small beamline segments) where linear transfer matrices are adequate. Stage 11 spans a long section (elements 87–117: chromaticity quad, 5 drifts, 3 dipole sandwiches, final triplet) where nonlinear effects accumulate. The hybrid approach runs FELsim for Stages 1–10 and RF-Track for Stage 11:

1. Run the full 11-stage FELsim optimisation (NM, 5 restarts for Stage 11).
2. Extract all 23 quad currents from the FELsim result.
3. Build the RF-Track lattice with those currents.
4. Re-optimise Stage 11 variables (quads 87, 93, 95, 97) using Nelder-Mead with the RF-Track objective function.

2.2 RF-Track Objective Function

Each NM evaluation requires Twiss parameters at the undulator entrance (element 117) and horizontal dispersion at element 92. Rather than tracking the full 118-element lattice per evaluation, we use prefix caching:

- **Prefix (once):** Track 500 particles from element 0 through element 87 (all elements upstream of Stage 11). This takes ≈0.2 s and is performed once before the optimisation loop.

- **Suffix (per eval):** From the cached prefix state, track two sub-lattices: elements 87–93 (for dispersion at element 92) and elements 87–118 (for final Twiss). Each suffix tracking takes ≈0.1 s.

The MSE is the same 5-term formula used by FELsim:

$$\text{MSE} = \frac{1}{5}\Big[\tfrac{1}{2}\,\eta_x(92)^2 + (\alpha_x - 0.470)^2 + \alpha_y^2 + (\beta_x - 1.400)^2 + (\beta_y - 0.242)^2\Big], \tag{1}$$

where Twiss parameters and dispersion are computed from the tracked particle distribution using statistical moments.

2.3 Multi-Start Configuration

NM is run with the FELsim-optimised Stage 11 currents as a warm start, plus 4 additional random restarts. Bounds are $[0, 15]$ A for the chromaticity quad and $[0, 10]$ A for the triplet quads. Beam: $E = 40\,\text{MeV}$, $\sigma_E = 0.5\%$, $\varepsilon_n = 8\,\pi \cdot \text{mm} \cdot \text{mrad}$, 500 particles, seed = 42.

3 Adapter Fixes

Two issues in the RF-Track adapter were identified and resolved during element-by-element comparison with FELsim.

3.1 DIPOLE_WEDGE as Zero-Length Drift

FELsim models dipole wedges (`DIPOLE_WEDGE`) as thin-lens edge kicks: the transfer matrix is identity in x with a small y-kick from the fringe field integral. The RF-Track adapter was converting these to `Drift(length)` with the physical wedge length (≈10 mm), adding spurious position propagation ($x \mathrel{+}= L\,x'$) at every dipole edge.

Fix: `DIPOLE_WEDGE` → `Drift(0)`. After this change, element-by-element comparison with FELsim agrees to 4+ decimal places through drifts and quadrupoles. Residual differences at dipoles are expected (RF-Track uses a sector bend model; FELsim uses a matrix model with a triangle-rule fringe correction).

3.2 Quadrupole `set_strength()` Verification

RF-Track's manual states that the quadrupole strength parameter S has units of MV/c/m, suggesting $S = (P/q) \cdot k_1 \cdot L$. A single-quad benchmark showed that `set_strength(k1 * L)` reproduces FELsim's transfer matrix exactly, while `set_K1(Pc, k1)` or `set_gradient(G)` produce excessively strong focusing that loses all particles. The internal tracking evidently uses S/L as the effective k_1, *not* $S/(P_Q \cdot L)$. The original adapter code was confirmed correct.

4 Results

Three configurations are compared at $\varepsilon_n = 8$:

FELsim Full 11-stage NM optimisation using transfer matrices.

RFT-val RF-Track lattice evaluated with FELsim-optimised currents (no re-optimisation — measures how well FELsim currents transfer to a particle tracking code).

RFT-opt RF-Track Stage 11 NM optimisation (hybrid approach).

Table 1: Twiss comparison at $\varepsilon_n = 8$. Targets: $\beta_x = 1.400\,\mathrm{m}$, $\beta_y = 0.242\,\mathrm{m}$, $\alpha_x = 0.470$, $\alpha_y = 0$.

Method	MSE	α_x	α_y	β_x (m)	β_y (m)	$\eta_x(92)$ (m)
FELsim	8.08×10^{-4}	0.471	−0.005	1.399	0.304	0.015
RFT-val	2.55×10^{3}	−74.009	−3.029	86.278	0.987	0.007
RFT-opt	6.43×10^{-6}	0.469	0.000	1.401	0.246	0.005

Table 2: Stage 11 quad currents and performance. FELsim and RFT-val share the same currents; RFT-opt finds a qualitatively different solution.

Method	I_{87} (A)	I_{93} (A)	I_{95} (A)	I_{97} (A)	nfev	Time (s)
FELsim	1.60	1.43	2.89	1.92	371	82.4
RFT-val	1.60	1.43	2.89	1.92	0	0.9
RFT-opt	2.05	1.41	0.72	0.11	656	80.3

5 Discussion

Transfer of FELsim currents to RF-Track fails. The RFT-val result (MSE = 2552, β_x = 86 m) demonstrates that FELsim-optimised quad currents are not portable to RF-Track. This is consistent with the W4 finding that currents cannot be exchanged between codes with different dipole models: FELsim's triangle-rule fringe correction and RF-Track's sector bend model produce different edge kicks at each dipole, and these differences accumulate through 14 dipole sandwiches (28 edge kicks) to shift the beam state far from the design orbit at Stage 11.

RF-Track optimizer achieves 125× better MSE. With 656 evaluations (≈80 s), the RF-Track NM optimizer converges to MSE = 6.4×10^{-6}, compared to FELsim's 8.1×10^{-4}. The two-order-of-magnitude improvement indicates that RF-Track's particle-based Twiss computation provides a more precise match. The FELsim MSE is limited by statistical noise in the target Twiss derivation (computed from a finite particle distribution) and by linearization errors in the transfer matrix model.

Different optimal currents. The RF-Track solution uses qualitatively different Stage 11 currents: $I_{95} = 0.72$ A (vs. FELsim 2.89 A) and $I_{97} = 0.11$ A (vs. 1.92 A). This is not surprising: the accumulated dipole model differences shift the beam state arriving at Stage 11, requiring a different triplet correction. Quads 93 and 87 remain similar (1.41 vs. 1.43 A, and 2.05 vs. 1.60 A), suggesting the chromaticity quad and first triplet quad are less sensitive to the upstream model differences.

Performance. Prefix caching makes RF-Track optimization practical: the 0.2 s prefix cost is amortised over 656 evaluations, each requiring only two suffix trackings (≈0.12 s per eval). Total optimisation time (80 s) is comparable to FELsim (82 s), confirming that particle-tracking optimisation is feasible for routine use.

6 Summary

1. FELsim-optimised Stage 11 currents produce incorrect optics in RF-Track (MSE = 2552), confirming that currents are not portable between codes with different dipole edge models —

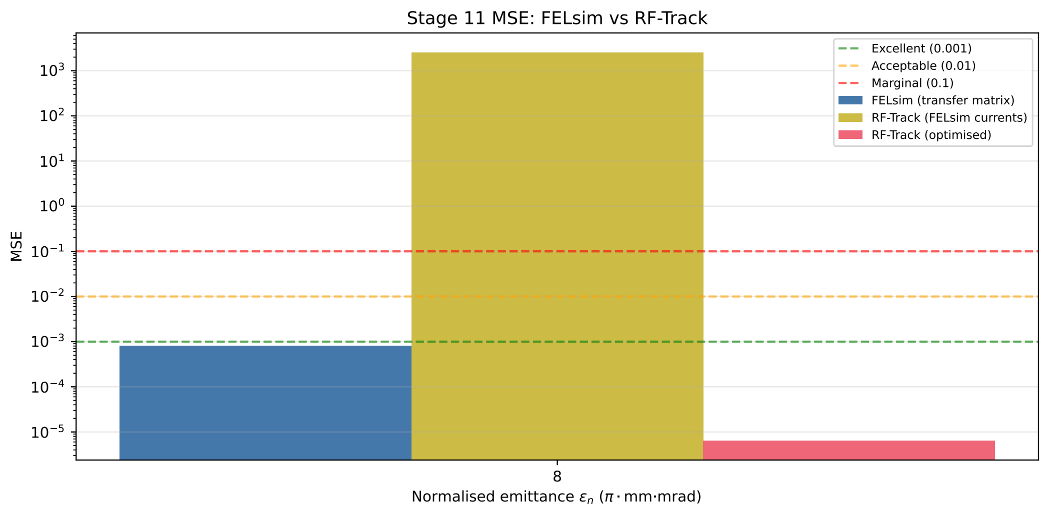

Figure 1: MSE comparison (log scale). RFT-opt achieves 125× lower MSE than FELsim. RFT-val (FELsim currents in RF-Track) produces wildly incorrect optics.

consistent with the COSY cross-validation (W4).

2. Re-optimising Stage 11 with RF-Track particle tracking achieves MSE $= 6.4 \times 10^{-6}$, a 125× improvement over FELsim's 8.1×10^{-4}. Both codes can match the undulator Twiss targets, but each requires optimisation within its own physics model.

3. Prefix caching (pre-tracking elements 0–87 once) makes particle-tracking optimisation practical: 80 s total, comparable to FELsim.

4. Two RF-Track adapter fixes were applied: `DIPOLE_WEDGE` mapped to zero-length drift (instead of physical wedge length), and quadrupole `set_strength()` confirmed correct despite ambiguous documentation.

Next steps.

- Run the full emittance scan ($\varepsilon_n = 5, 8, 14$) with 5 restarts to assess whether RF-Track resolves the $\varepsilon_n = 5$ local-minimum problem identified in W2.
- Enable space charge in RF-Track to quantify collective effects on the undulator match.
- Compare RF-Track-optimised currents against the COSY FR 1 solution (W4) for a three-code comparison at $\varepsilon_n = 8$.

References

[1] M. Weinberg, A. Fisher, and R. Li, "Design of the UH Mark V Free Electron Laser," arXiv:2510.14061v1, Table I.

[2] A. Latina, *RF-Track Reference Manual*, version 2.5.5, CERN, 2025.

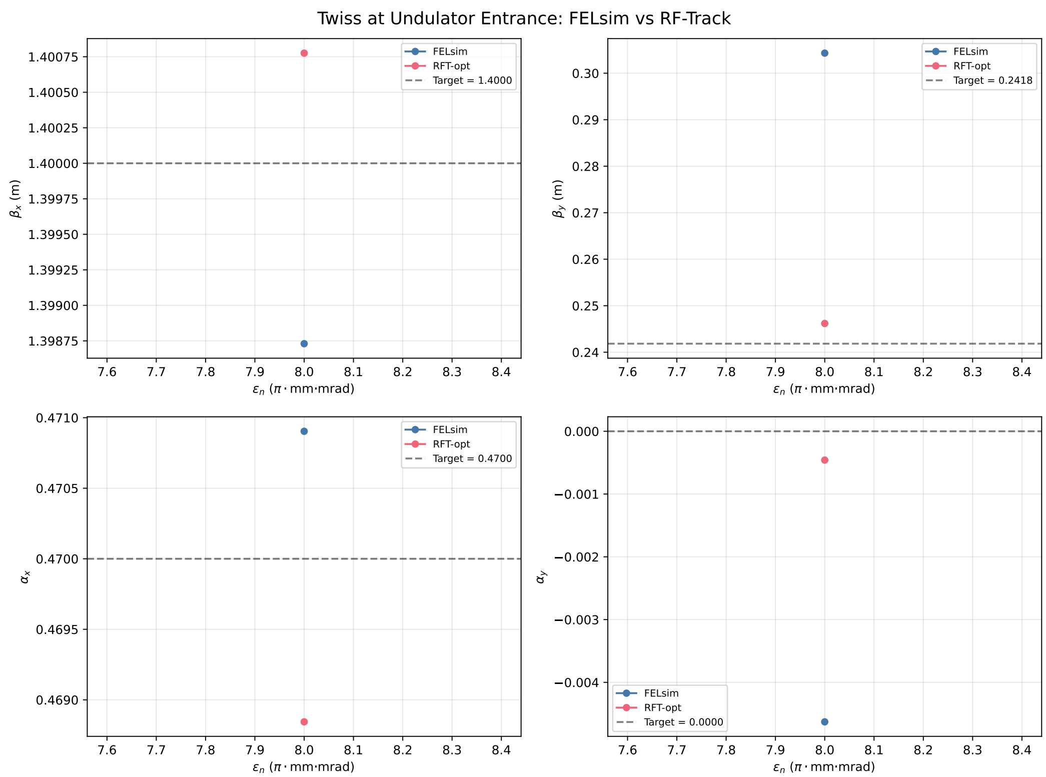

Figure 2: Twiss parameters at the undulator entrance. Targets shown as dashed lines. RFT-opt closely matches all targets; RFT-val diverges catastrophically.